

A socio-economic theory of regulatory compliance

Jon G. Sutinen

*Department of Environmental and Natural Resource Economics,
University of Rhode Island, Kingston, USA, and*

K. Kuperan

*Department of Natural Resource Economics,
Universiti Pertanian Malaysia, Malaysia*

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Abstract *An enriched theoretical model of regulatory compliance is developed in this paper. The body of empirical evidence demonstrates that the pure deterrence model of regulatory compliance, which focuses primarily on the certainty and severity of sanctions as key determinants of compliance, provides only a partial explanation of compliance behavior. To offer a more complete explanation, the model developed herein integrates economic theory with theories from psychology and sociology to account for both tangible and intangible motivations influencing individuals' decisions whether to comply with a given set of regulations. Specifically, the model accounts for moral obligation and social influence in addition to the conventional costs and revenues associated with illegal behavior. While cast in a natural resource management context, the theory developed here is applicable to a variety of institutional conditions. The resulting framework enables the design and implementation of more efficient compliance and regulatory programs than was heretofore possible.*

Introduction

Despite the worldwide trend away from central planning and governmental intervention in the marketplace, regulation of economic activities remains a fact of life. Natural resource industries, because of the prevalence of externalities and public goods, inevitably become highly regulated. Even where a full set of property rights is defined to internalize externalities, it is necessary to control excluded users from infringing on those rights. Other industries where market structure or public safety and security are potential problems also are subject to an array of government restrictions.

Regulated economic agents are typically controlled through monitoring, surveillance, and enforcement. This control is commonly left to separate enforcement authorities and not systematically considered when developing regulatory policies. There is little or no recognition of how policies and the policy process may affect the extent of compliance with regulations. Policy

analysis and formulation frequently assume perfect compliance can be achieved at no cost. Yet, when things go wrong, as they often do, enforcement is cited as one of the principal reasons for failure, and more and better enforcement is demanded.

Buying more and better enforcement services, however, is not cheap. In natural resource management programs, for example, enforcement is frequently the most costly element, accounting for a quarter to over a half of all public expenditures. This raises questions of whether there are ways to improve the cost-effectiveness of traditional enforcement, and whether there are ways to secure compliance without heavy reliance on costly enforcement.

Prescribing compliance policy and institutional design requires a sound understanding of compliance behavior. Becker (1968) was the first to develop a formal theoretical framework for explaining criminal activity. Following Bentham (1789) and Smith (1759, 1776), Becker argued that criminals behave basically like other individuals in that they attempt to maximize utility subject to a budget constraint. In Becker's model, an individual commits a crime if the expected utility from committing the crime exceeds the utility from engaging in legitimate activity. Becker's framework became the launching pad for a series of studies on the economics of crime[1]. The basic deterrence framework used in these studies assumes that the threat of sanctions is the only policy mechanism available to improve compliance with regulations.

The basic deterrence model, however, has at least two important shortcomings: first, the model does not explain the available evidence very well and, second, the policy prescriptions of the model are not very practical. Low expected penalties do not always result in high levels of non-compliance; and prescriptions for more enforcement inputs and higher penalties are usually unfeasible or not cost-effective.

In an attempt to overcome these and other shortcomings, this paper presents an enriched model of compliance behavior in which rational individuals are driven by both intrinsic and extrinsic motivations (including, but not restricted to, wealth enhancement). The model integrates economic theory with theories from psychology and sociology to account for both tangible and intangible motivations influencing individuals' decisions whether to comply with a given set of regulations. Specifically, the model accounts for moral obligation and social influence in addition to the conventional costs and revenues associated with illegal behavior.

The neoclassical deterrence model

We begin by reviewing the basic model of deterrence in a commercial fishery context. Let t represent the amount of time engaged in a regulated fishing activity (e.g. when and where fished, use of gear, species caught, quantity caught), where t_0 is the time fishing legally and t_1 the time fishing illegally. The set of regulations (gear and catch restrictions, closed areas and seasons, etc.) is represented by the vector R . The state of the resource stock and other

productivity conditions when fishing legally and illegally is given by x_0 and x_1 respectively. The individual's net revenue before penalties (if caught violating the regulation) in a period is given by $r(t_0, t_1, x_0, x_1)$, where $r_{t_0} \geq 0$, $r_{t_1} < 0$ and $r_{x_i} > 0$, $i = 0, 1$ [2]. If caught violating the regulation and sanctioned, the penalty is given by the penalty schedule $f(t_1)$, where $f(0) = 0$, $f_{t_1} \geq 0$. The parameters determining the structure of the penalty structure are denoted by the vector $\bar{f}$. The probability of detection and sanction in a period is given by $p(t_1)$, where $p(0) = 0$, $p_{t_1} \geq 0$ [3]. The individual's utility function is given by $U^j[\pi^j(\bullet)]$, $j = n, c, o$, where n is the index representing not caught violating, c caught violating, and o indicates the individual complies with the regulation (i.e. $t_1 = 0$). Profits for the period in each case are given by

$$\pi^c = r(t_0, t_1, x_0, x_1) - f(t_1),$$

$$\pi^n = r(t_0, t_1, x_0, x_1),$$

$$\pi^o = r(t_0, 0, x_0, x_1).$$

Two conditions govern the individual's compliance behavior, a total and a marginal condition. The total condition is that the individual violates if and only if

$$\max p U^c[\pi^c] + (1-p) U^n[\pi^n] \geq \max U^o[\pi^o].$$

That is if, at the optimal $t_0^*, t_1^* > 0$,

$$U^n[\pi^n] - p \{U^n[\pi^n] - U^c[\pi^c]\} \geq U^o[\pi^o] \quad (1)$$

The individual violates only if the benefits less the expected costs of violating exceed or equal the benefits of compliance. Otherwise, the individual complies.

Enforcement policy and practice determine the probability of detection, p , and the penalty, f . A sufficiently large increase in p or f , or both, operates to deter the individual from violating the regulation. In a heterogeneous fishery, where the equality holds in (1) for marginal violators, any increase in p or f will reduce the number of violators. The number of violators determined by (1) is denoted by N .

Given (1) is satisfied, the marginal condition for each of the N violators which determines the extent of their violation is:

$$r_{t_1} = \{p f_{t_1} + (p_{t_1} / U_{t_1}^c)(U^n - U^c)\} \theta \quad (2)$$

where $\theta = [(1-p)(U_{\pi\pi}^n/U_{\pi\pi}^c) + p]^{-1} \geq 1$, and $U_{\pi\pi}^j$ is the marginal utility of income. The individual violates up to the point where the marginal profit of violating equals the expected marginal penalty augmented by the risk factor θ . Note that $\theta > 1$ iff $U_{\pi\pi}^j < 0$, i.e. the individual is risk averse.

If both the marginal penalty and marginal probability (f_{t_1} and p_{t_1}) are zero, (2) becomes r_{t_1} . That is, under such conditions there is no deterrence effect at the margin, since p and f would have no effect on the extent of violations for the committed violator. Increases in the penalty and/or probability would reduce total violations only by reducing the number of violators, and would have no effect on remaining violators' behavior. Only when one or both of f_{t_1} and p_{t_1} are positive does deterrence operate to reduce the extent of violations by inframarginal violators.

The optimal amount of time fishing illegally by each violator, t_1^* , determined by the marginal condition (2), is given in general by

$$t_1^* = t_1^*(R, p, \bar{f}, x_0, x_1).$$

A measure of aggregate violation activity, the total amount of illegal fishing time (e.g. the total number of user days a closed area regulation is violated) is given by

$$T_1 = T_1(R, p, \bar{f}, x_0, x_1) = \sum_{i=1}^N t_1^*,$$

where N is determined by (1) and t_1^* by (2). If M is the total number of fishermen in the fishery, then $M-N$ is the total number of compliers.

The framework developed in this section has straightforward prescriptions for policy. If the typical illegal gains are greater than the gains from legal fishing, the expected penalty should be large enough to offset the difference between legal and illegal gains. Since enforcement is costly, the probability should be kept low and penalties high (i.e. large enough for the product of their monetary value with the low probability to be larger than the difference between legal and illegal gains).

In practice the probability usually is low. The typical odds of being caught violating a fishery regulation are below 1 percent, and often at or near zero (Bean, 1990; Furlong, 1991; Sutinen and Gauvin, 1988). Penalties, on the other hand, generally are not large enough. For example, in the groundfish fishery of the northeast USA, Sutinen *et al.* (1990) estimated flagrant violators grossed about US\$15,000 per trip from violating closed area and mesh size regulation. During 1987 this resulted in illegal earnings of \$225,000 for flagrant violators. Typical penalties when caught and sanctioned for these violations ranged from \$3,000-\$15,000 in monetary fines. A similar pattern of potential illegal gains relative to the certainty and severity of sanctions tends to appear in most fisheries.

High enough penalties are generally not feasible. The courts are not willing to mete out sanctions perceived as excessively severe. Rather, courts tend to impose sanctions that fit the crime, as measured by the illegal gains realized or the social harm caused. Hence, the sanctions for violations of fishing regulations will generally be modest and, according to the basic deterrent framework, not act as an adequate deterrent to illegal fishing. Yet, despite this apparent weakness, a high proportion (half to 90 percent) of fishermen

normally comply with regulations (Bean, 1990; Sutinen and Gauvin, 1988; Sutinen *et al.*, 1990).

The task before us is clear. The framework must be strengthened to better explain the available evidence on compliance. Only then can we confidently prescribe superior cost-effective compliance policies.

Intrinsic motivation and morality

Asked why they persist to comply when illegal gains are so much larger than the expected penalties, many fishermen refer to the need to “do the right thing.” That is, they express an obligation to obey a set of values (either their own or an authority’s). The sense of moral obligation, it turns out, is very common throughout society and, it appears, may be a significant motivation explaining much of the evidence on compliance behavior. We now turn to incorporating intrinsic motivation into the basic deterrent model.

The paradigm commonly used in economics to explain and predict behavior (especially the theory used for policy analysis) makes little allowance for personal moral values. Most contemporary economic theory typically either ignores the influence of moral considerations or, in the extreme, denies moral factors have an influence on economic behavior (Hausman and McPherson, 1993, 1996). This raises the question whether many aspects of regulatory policy are ill founded, or whether the theory is incomplete.

Morality and moral norms clearly influence economic outcomes. The evidence supporting this proposition is substantial[4]. For example, a large number of experiments have shown that people do not act as free riders when given the opportunity. Instead, many people persist in investing substantial proportions of their resources into public goods despite conditions designed to maximize free riding. In experiments of repeated prisoner dilemma games, over half of the subjects cooperate without being coerced or paid. Several experiments have shown that many people return lost wallets to their owners with all of the money inside. We also witness anonymous contributions to charity – above and beyond what tax incentives can explain. It is customary for people to leave tips in restaurants in distant cities which they never expect to visit again. And, as noted above, many fishermen comply despite large potential illegal gains and small expected penalties.

In contrast to contemporary economists, our economist forefathers gave morality due attention. Smith (1759) explicitly portrays human economic motivation to be multidimensional, arguing that psychic wellbeing is based on acting morally and receiving the approval of others, as well as enhancing wealth. For the intrinsic motivation influencing behavior, he imagines an “impartial spectator” within each of us, with which we “scrutinize the propriety of our own conduct.” Nineteenth century economists commonly account for moral sentiments in their writings. Contemporary economists, of course, do not completely ignore morality. Some prominent figures in the profession, including Akerloff, Harsanyi, Hirshman, Margolis, Schelling, Sen and Stigler, have examined the consequences for economic outcomes of morality (see Hausman

and McPherson (1993, 1996) for an excellent survey of recent literature). The fact remains, however, that morality is a foreign element to most contemporary economic analysis. A socio-economic theory

Outside economics, recent research in psychology and sociology emphasizes the importance of socialization processes in affecting behavior. Compliance with rules and regulations is hypothesized to be related to both the internal capacities of the individual and external influences of the environment, where the socialization process is the linkage between the individual and society. There are two leading psychological theories to explain how socialization processes work with respect to compliance behavior: cognitive and social learning. Cognitive theory focuses primarily on the individual and stages of development (cf. Kohlberg, 1969, 1984; Levine and Tapp, 1977; Tapp and Kohlberg, 1977). According to cognitive theory, the key variables determining compliance are the individual's personal morality and level of moral development.

Social learning theory, on the other hand, focuses primarily on the conditioning effects of the environment (cf. Akers, 1985; Akers *et al.*, 1979; Aronfreed, 1968, 1969; Bandura, 1969; Mischel and Mischel, 1976). According to social learning theory, the key variables determining compliance include peers' opinions, and the extent of social influence an individual encounters.

The sociology literature contains two basic perspectives on compliance: instrumental and normative (cf. Tyler, 1990). Similar to Becker, the instrumental perspective assumes individuals are driven purely by self-interest and respond to changes in the tangible, immediate incentives and penalties associated with an act. The key variables determining compliance are the severity and certainty of sanctions.

The normative perspective emphasizes what individuals consider just and moral, instead of what is in their self-interest. Individuals tend to comply with the law to the extent that they perceive the law as appropriate and consistent with their internalized norms. The key variables determining compliance in the normative perspective are individuals' perceptions of the fairness and appropriateness of the law and its institutions.

In summary, the literature identifies the following factors determining compliance: potential illegal gain, severity and certainty of sanctions, individuals' moral development and their standards of personal morality, individuals' perceptions of how just and moral are rules being enforced, and social environmental influences[5]. Next, we develop a theoretical model consistent both with this literature and with basic principles of economics.

Our theoretical development adopts the perspective of Adam Smith on individual behavior. To incorporate intrinsic motivation in the model, we imagine Smith's "impartial spectator" operating within the individual to scrutinize and evaluate the propriety of his/her conduct[6]. Since proper conduct depends on whether an individual acts morally, we must characterize the nature of moral acts.

While there is no universal definition of moral acts, Etzioni (1988) identifies several characteristics which are generally agreed on. One of the primary characteristics of moral acts is that they are motivated intrinsically, involving non-material rewards internal to oneself. That is, internal satisfaction is realized independent of extrinsic consequences, such as whether others know about such behavior. A second characteristic of moral acts is that sacrifice and the denial of pleasure (e.g. doing penance, fasting) in the name of moral principle are often involved. An implication of this is that individuals will sacrifice income or incur costs to carry out a moral act. A third characteristic is that moral acts often concern intentions and processes, not outcomes. Unlike consumptive pleasure, moral satisfaction can be the result of taking proper measures, regardless of the outcome. To the extent that moral acts are concerned with the end results, how the result was attained is significant. For example, stealing a candle to light in church would not be morally satisfying to most people. A fourth characteristic is symmetry. That is, the standard defining morality is applied equally to all people under comparable circumstances. Otherwise the moral dictum is arbitrary. Each of these characteristics has significant implications for our model of compliance behavior and, it turns out, for compliance policy.

Let $m(t_1)$ represent the level of the individual's personal moral standing. If the individual complies with the regulation then the level of personal moral standing is given by $m(0)$. If violating the regulation is contrary to the individual's values, then for $m_{t_1} < 0$ for $t_1 > 0$ [7], and $m(0) > m(t_1)$ when $t_1 > 0$. The utility function of the individual is modified to include intrinsic motivation as follows:

$$U^j[\pi^j(\bullet), m(t_1)], \text{ where } U_m^j > 0.$$

The total condition on violation becomes, at $t_0^*, t_1^* > 0$,

$$U^n[\pi^n, m] - p\{U^n[\pi^n, m] - U^c[\pi^c, m]\} \geq U^o[\pi^o, m] \quad (3)$$

and the marginal condition is

$$r_{t_1} = \{pf_{t_1} + (p_{t_1} / U_{t_1}^c)(U^n - U^c)\}\theta + m_{t_1}\sigma \quad (4)$$

where

$$\sigma = -[(1-p)U_m^n + pU_m^c] / [(1-p)(U_\pi^n / U_\pi^c) + p] < 0.$$

It is clear from (3) and (4) that the presence of intrinsic motivation, when violating the regulation is contrary to the individual's moral values, reduces violations. The stronger individuals' personal moral standing is negatively affected, the fewer violators and the less violation by each of the inframarginal violators. Furthermore, one can imagine a not too unrealistic setting in which $p = 0$ and yet a high degree of compliance is realized.

Extrinsic motivation

In addition to wealth enhancement, social influence is another extrinsic motivation influencing behavior. Concern for one's social reputation has long been recognized as a motivation important to compliance behavior (c.f. Allingham and Sandmo, 1972)[8]. Social influence and morality are closely linked. The symmetry characteristic of moral acts implies that the standards used to judge one's own behavior are used to judge others' behavior. Therefore, the moral principles on which individuals base their own behavior are also the basis for the social influence they exercise. Social influence to conform is expected to be stronger the more widespread a common moral obligation in the fishing population.

Social influence plays a significant role in everyday social exchange, often taking the subtle forms of ostracism or withholding of favors. Like enforcement authorities, peer groups can reward and punish their members, either by withholding or conferring signs of groups status and respect, or more directly by channeling material resources toward or away from a member of the group. The available evidence supports the hypothesis, showing that a given individual is more non-compliant the more his community and peer groups are non-compliant (Geerken and Gove, 1975; Vogel, 1974; Witte and Woodbury, 1985). Social influence in fisheries is often manifested in forms of verbal and physical abuse (e.g. fist fights, destruction of gear and vessels). In the Massachusetts lobster fishery strong forms of social influence, commonly called "self-enforcement," are estimated to account for the bulk of enforcement in the fishery (Sutinen and Gauvin, 1988). Fisheries with which we are familiar where social influence to comply with management regulations is prevalent, and appears highly forceful, include American lobster (Massachusetts and Maine), clam (Rhode Island), herring roe (Alaska, British Columbia, Oregon, San Francisco Bay), sakuri ebe (Japan). There probably are many other fisheries where this phenomenon is operative.

To incorporate social influence as a form of extrinsic motivation in the utility function let $S(t_1)$ represent the level of the individual's social standing, where, if violating the regulation is against the social norm, $S_{t_1} < 0$ for $t_1 > 0$. Assume, for present purposes, that the individual's social reputation is affected only if detected and sanctioned by enforcement authorities[9]. Therefore, if not detected the individual's social standing remains at $S(0)$. The utility function of the individual now includes this as an argument:

$$U^j[\pi^j(\bullet), m(t_1), S(t_1)], \text{ where } U_S^j > 0.$$

The total condition on violation becomes, at $t_0^*, t_1^* > 0$,

$$U^n[\pi^n, m, S] - p\{U^n[\pi^n, m, S] - U^c[\pi^c, m, S]\} \geq U^o[\pi^o, m, S] \quad (5)$$

and the marginal condition is

$$r_{it} = \left\{ p f_{it} + \left(p_{it} / U_{it}^c \right) (U^n - U^c) \right\} \theta + m_{it} \sigma + S_{it} \mu \quad (6)$$

where

$$\mu = -[p U_s^c] / [(1-p)(U_\pi^n / U_\pi^c) + p] < 0.$$

From (5) and (6) it is clear that the presence of social influence, when violating, is contrary to social norms, reduces the number of violators and the extent of violations by each violator. As with personal moral standing, the stronger a person's social standing is negatively affected, the fewer the violators and the less violation by each of the inframarginal violators. Combined, moral obligation and social influence potentially can generate significant levels of compliance even in the face of a weak deterrent effect.

Morality, fairness and suasion

We now have an improved framework to explain how an individual's and the community's norms of behavior influence compliance. As it stands, the framework is not very useful for policy analysis since the norms of behavior seem to be exogenously determined and given. There is evidence to suggest, however, that individuals' moral and social standing may be closely linked to regulatory policy and practices.

The compliance literature recognizes two types of intrinsic motivation or obligation (Tyler, 1990). One is related to the individual's desire to behave according to his sense of personal morality, i.e. an internal obligation to follow one's own sense of what is right or wrong. The other type is related to the intrinsic obligation to follow the dictates a "legitimate" authority, such as the police, one's boss, or other authority. Legitimacy effectively functions as a stock of loyalty on which leaders can draw. Those who accept an authority's legitimacy are expected to comply with its dictates even when the dictates are contrary to an individual's self-interest[10]. These components of intrinsic motivation, and their determinants, are illustrated in Figure 1.

Legitimacy

The theories in the compliance literature identify four sets of an authority's characteristics which relate to legitimacy. Two involve outcomes (or ends), and two involve processes (or means) of the authority; and two involve issues of justice, and two do not. The first of these, effectiveness of the outcome, may involve the extent to which conservation is realized and an individual fisherman is made better off. The second, distributive justice of the outcome, involves the perceived fairness of how the benefits or sacrifices are shared among the affected parties. The third, efficiency of the process, involves the speed and efficiency with which people perceive the authority responding to problems within the scope of the authority's jurisdiction. The fourth, procedural

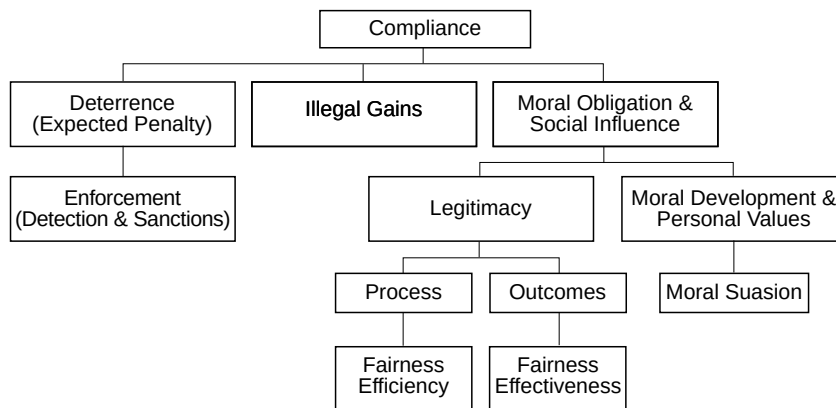


Figure 1.
Determinants of
compliance

justice, involves how fairly the authority treats people and the concerns of those affected by the process.

Public choice theory in general, and some psychological theories of leadership, argue that legitimacy depends in large part on the authority's ability to provide favorable outcomes. That is, people perceive as legitimate and obey the institutions that produce positive outcomes for them. However, other evidence indicates people place great importance on procedural issues. The principal result of Tyler's (1990) study is that the perception of legitimacy is closely linked to people's views of the fairness of the procedures used by the authorities. More strongly, he demonstrates that the people he studied comply more with the law if the procedures employed by the legal or political authority are perceived by them to be fair[11]. Other research cited by Tyler indicates that distributive justice is another critical determinant of legitimacy and compliance. Tyler's overall conclusion is that justice matters most, with procedural justice being more important than distributive justice, and the efficiency of the process and favorability of the outcome matter less in promoting legitimacy.

To incorporate the factors of moral development and legitimacy, the function representing the individual's level of personal moral standing is expanded to $m(t_1, h, l)$, where h represents the individual's moral development (e.g. using Kolberg's scale), and l represents the legitimacy of the regulation as perceived by the individual. Legitimacy, l , is a vector of Tyler's process and outcome variables. A violation is assumed to reduce an individual's moral standing more the higher either h or l , i.e. $m_{t_1 h} < 0$, $m_{t_1 l} < 0$ for all $t_1 > 0$.

Moral development and personal values

Social psychology emphasizes the importance of an individual's personal characteristics in determining compliance behavior. Moral development of the individual is hypothesized to be directly related to one's propensity to comply with society's rules. Kohlberg (1969, 1976, 1981, 1984) argues there are three distinct levels of moral development: pre-conventional, conventional, and post-

conventional. Each level characterizes the relationship between the individual and his social environment, based on the individual's attitude toward society's conventions. Pre-conventionals tend to reason in terms of fear of punishment rather than in terms of the harmfulness of their acts or the need for social order. Conventionalists tend to reason in terms of social conformity and order. Post-conventionalists tend to reason in terms of moral principles that are independent of social order. Kohlberg (1969, 1976, 1984) argues that rule violations tend to decrease at higher levels of moral development. The available evidence supports this hypothesis[12].

It is generally argued that individuals' personal values, or norms, influence their compliance behavior[13]. While there is substantial evidence that personal values influence behavior and economic outcomes (cited above), the correspondence between personal values and compliance with the law is mixed. Personal moral values may or may not accord with the law and dictates of an authority, leading to resistance to the law and to legal authority instead of compliance (Tyler, 1990). For example, Westat (1980) reports that taxpayers are generally ambivalent about cheating on their tax return, especially when small amounts are concerned.

Where personal moral values do not adequately support policy objectives, education can provide people with information which, if successful, shapes their attitudes and values, and induces socially desirable behavior. The general operating hypothesis is that information functions as a form of moral suasion, influencing personal values, which in turn influence behavior (Fishbein and Azjen, 1975).

Aggregate compliance behavior and equilibrium

The behavior of others (Sutinen *et al.*, 1990; Young, 1979) is expected to affect individual compliance via the nature and extent of social influence exerted in the community. Since the moral principles on which individuals base their own behavior are the same basis for the social influence they exercise, social influence depends on the larger community's moral development and perception of the institution's legitimacy. A reasonable inference is that the extent of noncompliance in the fishing community reflects the amount of social influence exerted within the community. Therefore, the social influence function is extended to include V , the perceived amount of noncompliance in the fishery, i.e. $S(t_1, V)$.

In the presence of both moral obligation and social influence the optimal amount of time fishing illegally by each violator, t_1^* , determined by marginal condition (6), is given in general by

$$t_1^* = t_1^*(R, p, \bar{f}, x_0, x_1, h, l, V). \quad (7)$$

As explained below, the presence of V in (7) has some important implications for the analysis of compliance behavior and policy.

The behavior of others is expected to affect individual compliance behavior for game theoretic reasons as well. Individuals face an “assurance problem,” where the payoff from complying depends on the extent of assurance that others also will comply (Sen, 1967). The higher the compliance by others, the stronger the individual’s incentive to comply (Runge, 1981, 1984). That is, the extent of compliance by others may also affect the total condition, (5), when cast in a strategic context.

Our measure of aggregate violation activity, the total amount of illegal fishing time, is given in general terms by the expression

$$T_I = T_I(R, p, \bar{f}, x_0, x_I, h, t, V) = \sum_{i=1}^N t_i^* , \quad (8)$$

where N is determined by the total condition (5), and t_1^* by the marginal condition (6). While this expression for aggregate behavior is similar to that for the basic deterrence model, the presence of V implies different behavioral properties in equilibrium.

Since the amount of illegal fishing time by any one fisherman is directly related to that of all other fishermen (all being subject to the same influences), then t_1 and V are directly related. And, there is a single equilibrium t_1 for each level of V , and vice versa. This means that t_1^* in (7) is more elastic in any of the policy variables than in the basic case. Consider an increase in the penalty. For a given level of V the increased penalty decreases t_1 as in the basic case. Since t_1 is reduced for all violating fishermen V is reduced, further reducing t_1 for each violator. There is, in effect, a multiplier effect or, what has come to be known in the compliance literature, a general deterrence effect. That is, an enforcement action that reduces one individual’s violation activity (a specific deterrence effect) also reduces the violation activity of other individuals (a general deterrence effect). We expect this general deterrence effect to be greater the more flagrant and chronic the violations of the individual targeted for enforcement action.

Testable hypotheses of compliance behavior

The model is capable of explaining or predicting the effects on:

- the number of violators;
- the amount of noncompliance by individuals; and
- the total amount of noncompliance when policies and other factors change.

This is investigated by differentiating the total and marginal conditions to generate a set of testable hypotheses (see the Appendix). The comparative statics results for the two conditions gives rise to the following testable hypotheses on the violation decision.

Both the number of violators and the amount of illegal fishing time by each violator will be reduced by any one, or combination of, the following changes:

- an increase in the probability of detection and sanction;
- an increase in the size of penalty;
- an increase in the level of the individual's moral development; or
- an increase in the legitimacy of the regulatory authority;
- a decrease in the size of the stock fished illegally; and
- a decrease in the amount of aggregate violations.

An increase in the legally fished stock can either increase or decrease the number of fishermen violating the regulation, and will have no effect on the amount of illegal fishing time by individual violators. These changes have no effect on the amount of legal fishing time, with one exception. An increase in the legally fished stock will increase the amount of legal fishing time by individual fishermen.

Concluding remarks

The basic deterrence model is extended above to allow individual's behavior to be driven by both intrinsic and extrinsic motivations. The resulting model integrates economic theory with theories from psychology and sociology to account for moral obligation and social influence in addition to the conventional costs and revenues associated with illegal behavior. The extended model explains how existing compliance policies are expected to work to control illegal activity. Not only is the model consistent with actual policy approaches, it is consistent with received theory and supported by a wide body of evidence on human behavior.

The willingness to comply stemming from moral obligation and social influence is based, *inter alia*, on the perceived legitimacy of the authorities charged with implementing the regulations. Some evidence suggests that a key determinant of perceived legitimacy is the fairness built into the procedures used to develop and implement policy. To the extent that this is valid, regulatory authorities should determine what policies and practices are judged fair by those segments of the population subject to regulations. This may mean, for example, that civil penalties and other sanctions should be comparable in value to the larger of the harm done or gains realized. This may mean that individuals subject to surveillance and monitoring be treated with dignity and respect. This may mean that regulations must appear reasonable and "make sense."

To the extent that the theory is supported by empirical evidence, there are a number of conclusions for policy that flow from the model developed here. Perhaps the most important implication is that top-down, command and control-style policies likely will not be perceived as legitimate. The result will be policy that is ineffective in achieving its goals, and a program that is costly and rife with popular dissatisfaction.

Another implication of the theory is that policy makers should pay more attention to the fundamental issues of institutional design. Like other political

and legal institutions in society, regulatory bodies should devote great effort to developing legitimacy. The mere fact that an institution is formed under a piece of legislation does not necessarily confer on it legitimacy. How legitimacy can be earned is beyond the scope of this short contribution, but is an important issue worthy of future investigation.

A socio-economic
theory

More equitable procedures for imposing restrictions on the economic community should strengthen legitimacy and voluntary compliance, according to this theory. Co-management regimes, in which participants are empowered to play a prominent role in decision making, may be a means of achieving this end (Hanna, 1995). This would address, in part, the need to incorporate procedural justice in the institution. Similarly, enforcement policies and practices need to be seen by participants to be fair. This may mean (though we do not know) that enforcement authorities should target chronic and flagrant violators of the regulations, punishing them accordingly, while tolerating to some degree minor violations by individuals who normally comply with the regulations.

187

To the extent that this theory has merit, a legitimate regulatory authority will possess a precious stock of loyalty on which it can draw in times of crisis. A collapse of a fishery resource, or environmental calamity, can be swiftly and severely dealt with by a legitimate authority, imposing on participants significant short-term sacrifices. Participants who view the authority as legitimate feel a strong obligation to comply even when the dictates of the authority are contrary to their self-interest.

It seems that efficiency and equity may be complements instead of substitutes, at least in the context of institutional design and implementation. Also note that procedural equity appears to be more important than outcome equity. This suggests that we can still aim for economically efficient outcomes, viz. the allocation of resources, etc.

Coercive enforcement measures remain an essential ingredient in any compliance regime, even where a high degree of compliance is realized via the twin forces of moral obligation and social influence. As noted above, in almost any group of individuals subject to regulation there is a core subgroup (usually small) of chronic, flagrant violators motivated largely by the direct tangible consequences of their actions. Moral obligation and social influence have little or no effect on their behavior. Only by changing the economic incentives, by reducing the potential illegal gains or by increasing the expected penalty, can their illegal activity be controlled. In the absence of incentive programs, the only control mechanism for this subgroup is enforcement.

Even if the subgroup of chronic, flagrant violators is small and the amount of their illegal activity is minor, they need to be controlled. Otherwise, flagrant violators would appear to flaunt their violation of the law and to be immune to the regulations. This sends two signals to normally law-abiding participants. One is that regulatory procedures are unfair, having no effect on flagrant violators. The other is that the regulatory program is not effectively achieving its purpose (e.g. protecting the fishery resource). Each of these signals weakens the moral obligation to comply and the moral basis on which social influence is

exercised. As moral obligation and social influence are weakened, compliance begins to erode among those who would normally comply with the regulations. Their subsequent noncompliant behavior influences others not to comply with the regulations, and ultimately compliance breaks down[14]. Only effective enforcement can reverse and prevent this undesirable outcome.

Notes

1. See Heineke (1978) and Pyle (1983) for an overview of the theoretical models used in the economic literature of criminal behavior. More recently, Sutinen and Andersen (1985), followed by Anderson and Lee (1986) and Milliman (1986), combined Becker's deterrence model with a bioeconomic model to investigate various aspects of fisheries law enforcement. All address the issue of optimal quantities of enforcement services and management policies.
2. Subscripts of variables denote partial derivatives.
3. The probability is also a function of enforcement resources and avoidance inputs used by the individual, taken as fixed for the present analysis. Also note that $p(t_1)$ and $f(t_1)$ are not necessarily continuous functions.
4. For reviews of the evidence see Etzioni (1988, Chapter 4), Frank (1988), Mansbridge (1990), and Thaler (1991).
5. Which of these variables are significant determinants of compliance with regulations is ultimately an empirical issue.
6. How morality should be modeled is not without controversy. In our view, how it should be modeled is ultimately an empirical issue.
7. As with the probability and the penalty functions, the personal moral standing function, $m(t_1)$, is not necessarily a continuous function of t_1 . For example, $m(t_1)$ may simply shift to a lower level for any positive value of t_1 .
8. For a rigorous and comprehensive analysis of social exchange, social norms, and social influence, see Coleman (1990). Also see Ajzen and Fishbein (1980), Fishbein and Ajzen (1975), Muller (1979), Hoffman (1977) and Wrong (1980).
9. In most fisheries, however, a fisherman typically knows whether another fisherman is violating. In this case $S(x)$ functions similarly to $m(x)$.
10. Tyler argues that it is better for a rule-making body to base compliance on legitimacy than on personal or group morality because the scope of legitimate authority is more flexible (in that leaders usually have a wide range of discretionary authority). Personal morality, on the other hand, is double-edged, for it may or may not accord with the dictates of the authority, leading to resistance to the law and legal authority, instead of compliance with its dictates.
11. Fairness, justice, and equity, like morality, can have different meanings to different people, and defy precise definition for all situations. Tyler finds, however, that for a given situation or context, people's perceptions of these traits are remarkably consistent across the population of people affected by the authority and its rules and laws. For example, procedural justice often is associated with having neutral, honest authorities who allow people the opportunity to state their views, and authorities that treat people with dignity and respect.
12. Kuperan and Sutinen (1994) find that individual Malaysian fishermen are more compliant with fishery regulations, the higher the individual fisherman is on Kohlberg's scale of moral development.
13. Blasi (1980) notes there are substantial definitional, conceptual, and methodological obstacles to research on this issue. A few economists who have explored the theoretical

consequences of norms and moral values for economic behavior include Akerloff (1980, 1982, 1983), Frank (1985, 1987, 1988) and Schelling (1984).

14. This process of deteriorating compliance is believed to have occurred in Northeast fisheries in the late 1980s (Sutinen *et al.*, 1990).

A socio-economic
theory

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Appendix

Two sets of comparative statics results are presented here, one for the total condition governing the violation decision and thereby determining the number of violators, and the other for the marginal conditions governing the extent of violations.

Comparative statics results for total condition

As discussed above, the total condition specifies that the fishermen will violate if and only if the utility from fishing legally and illegally exceeds the utility from fishing legally only. Define:

$$G = \max_{\{t_0, t_1\}} p U^c [\pi^c(t_0, t_1, x_0, x_1), m(t_1, h, l), S(t_1, V)] + (1-p) U^n [\pi^n(t_0, t_1, x_0, x_1), m(t_1, h, l), S(t_1, V)],$$

$$Q = \max_{\{t_0\}} U^o [\pi^o(t_0), m(0, h, l), S(0, V)], \text{ and}$$

$$Y = G - Q.$$

Change in the probability of detection and sanction

$$\frac{dY}{dp} = [U^c - U^n] < 0, \text{ since } U^c < U^n$$

An increase in the probability of detection and sanction p will decrease the number of fishermen violating.

Change in the level of penalties (F)

$$\frac{dY}{df} = -pU_{\pi}^c \frac{df}{df} < 0$$

An increase in the size of penalties reduces the number of fishermen violating the regulation.

Change in the level of the illegal fish stock

$$\frac{dY}{dx_i} = \{ pU_{\pi}^c \left(\frac{d\pi^c}{dx_i} \right) + (1-p)U_{\pi}^n \left(\frac{d\pi^n}{dx_i} \right) \} > 0$$

An increase in the stocks fished illegally (such as in closed areas) will increase the number of fishermen violating the regulation.

Change in the level of the legal fish stock

$$\frac{dY}{dx_0} = pU_{\pi}^c \left(\frac{d\pi^c}{dx_0} \right) + (1-p)U_{\pi}^n \left(\frac{d\pi^n}{dx_0} \right) - U_{\pi}^o \left(\frac{d\pi^o}{dx_0} \right) \leq 0 \text{ or } \geq 0$$

An increase in the legally fished stock can either increase or decrease the number of fishermen violating the regulation.

Change in the level of moral development (h)

$$\frac{dY}{dh} = \{ pU_m^c + (1-p)U_m^n \} \frac{dm(t_i^*, h, l)}{dh} - U_m^o \frac{dm(0, h, l)}{dh} < 0,$$

assuming

$$U_m^c = U_m^n = U_m^o, \text{ i.e., } U_{m\pi}^j = 0, U_{mm}^j = 0.$$

An increase in the moral development level reduces the number of fishermen violating the regulation.

Change in the level of legitimacy (l)

$$\frac{dY}{dl} = \{ pU_m^c + (1-p)U_m^n \} \frac{dm(t_i^*, h, l)}{dl} - U_m^o \frac{dm(0, h, l)}{dl} < 0.$$

Greater legitimacy reduces the number of fishermen violating the regulation.

Change in the level of aggregate violations (V)

$$\text{Assuming } \frac{dS(0, V)}{dV} = 0, \text{ and } \frac{dS(t_i^*, V)}{dV} > \frac{dS(0, V)}{dV} \text{ for } t_i^* > 0 \text{ and any } V:$$

$$\frac{dY}{dV} = \{ pU_s^c + (1-p)U_s^n \} \frac{dS(t_i^*, V)}{dV} - U_s^o \frac{dS(0, V)}{dV} > 0$$

An increase in the level of aggregate violations increases the number of fishermen violating the regulation.

Comparative statics results for marginal conditions

The comparative statics results for the marginal condition are obtained by differentiating the first order conditions with respect to the variables of interest. The first order conditions are

$$V_0 = pU_{\pi}^c \pi_{t_0}^c + (1-p)U_{\pi}^n \pi_{t_0}^n = V_0(p, \bar{f}, t_0, t_1, x_0, x_1, h, l, V) = 0 \quad (A.1)$$

$$V_1 = p[U_{\pi}^c \pi_{t_1}^c + U_m^c m_{t_1} + U_S^c S_{t_1}] + (1-p)[U_{\pi}^n \pi_{t_1}^n + U_m^m m_{t_1} + U_S^m S_{t_1}] = V_1(p, \bar{f}, t_0, t_1, x_0, x_1, h, l, V) = 0 \quad (A.2)$$

Differentiating this system with respect to the parameters $p, \bar{f}, x_0, x_1, h, l, V$, and solving yields the following:

193

Effects on t_0	Effects on t_1
$\frac{dt_0}{dp} = \frac{-V_{0p}V_{11} + V_{1p}V_{01}}{[D]} = 0$	$\frac{dt_1}{dp} = \frac{-V_{1p}V_{00} + V_{0p}V_{10}}{[D]} < 0$
$\frac{dt_0}{d\bar{f}} = \frac{-V_{0\bar{f}}V_{11} + V_{1\bar{f}}V_{01}}{[D]} = 0$	$\frac{dt_1}{d\bar{f}} = \frac{-V_{1\bar{f}}V_{00} + V_{0\bar{f}}V_{10}}{[D]} < 0$
$\frac{dt_0}{dx_0} = \frac{-V_{0x_0}V_{11} + V_{01}V_{1x_0}}{[D]} > 0$	$\frac{dt_1}{dx_0} = \frac{-V_{1x_0}V_{00} + V_{10}V_{0x_0}}{[D]} = 0$
$\frac{dt_0}{dx_1} = \frac{-V_{0x_1}V_{11} + V_{01}V_{1x_1}}{[D]} = 0$	$\frac{dt_1}{dx_1} = \frac{-V_{1x_1}V_{00} + V_{10}V_{0x_1}}{[D]} > 0$
$\frac{dt_0}{dh} = \frac{-V_{0h}V_{11} + V_{01}V_{1h}}{[D]} = 0$	$\frac{dt_1}{dh} = \frac{-V_{1h}V_{00} + V_{10}V_{0h}}{[D]} < 0$
$\frac{dt_0}{dl} = \frac{-V_{0l}V_{11} + V_{01}V_{1l}}{[D]} = 0$	$\frac{dt_1}{dl} = \frac{-V_{1l}V_{00} + V_{10}V_{0l}}{[D]} < 0$
$\frac{dt_0}{dV} = \frac{-V_{0V}V_{11} + V_{01}V_{1V}}{[D]} = 0$	$\frac{dt_1}{dV} = \frac{-V_{1V}V_{00} + V_{10}V_{0V}}{[D]} > 0$